Dynamic Systems Inc.

CERTIFICATE OF CALIBRATION

Customer: University of Witwatersrand Serial No. 504

GLEEBLE SYSTEM CALIBRATION

CHECKLIST AND SUPPORT DOCUMENTATION

(for Model 1500, 1500D, 1500/20, 1000, 2000, 3500 and 3800 Gleeble Systems)

Company Name University of Witwatersrand

Gleeble Model Number 3500 Gleeble Serial Number 504

Room Temperature 22C

Calibration services were provided on the above listed Gleeble System in accordance with standard factory specified procedures. Calibrations performed have been recorded in this document.

This is to certify that the measurement system of the Gleeble listed has been calibrated using transfer standards traceable to the National Institute of Standards and Technology. The accuracy of the thermocouple wire is not included in this calibration process.

Accepted:

michael Bodunrin
(Customer)

Garrick Palermo
(Dynamic Systems Inc.)

3/2/2022
(Date)

Unless otherwise noted, calibration expires one year from above date.

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THERMOCOUPLE MEASUREMENT SYSTEM

REFERENCE INSTRUMENT	MODEL NUMBER	SERIAL NUMBER	CALIBRATION DUE DATE
Fluke	714	4141046	1/31/2023

DEGREES C X DEGREES F

REFERENCE TEMPERATURE INPUT		TC- 1 TYPE K			TC- 2 TYPE K		
DEG. C	DEG. F	AS FOUND	OOT	AFTER CALIBRATION	AS FOUND	OOT	AFTER CALIBRATION
200	400	200		200	200		200
300	600	300		300	300		300
400	800	399		399	400		400
500	1000	499		499	500		500
600	1200	599		599	600		600
700	1400	699		699	700		700
800	1600	799		799	800		800
900	1800	900		900	9000		9000
1000	2000	1000		1000	1000		1000
1100	2200	1099		1099	1100		1100
1200	2400	1200		1200	1200		1200
1300	2600	1299		1299	1300		1300
1400	2800	****		****	****		****
1500	3000	****		****	****		****
1600	3200	****		****	****		****
1700	3400	****		****	****		****

Note: Platinum Thermocouples - start @ 500°C (932°F)
Chromel-Alumel Thermocouples - start @ 200°C (392°F)

Date of Calibration: 2/28/2022

Calibrated by: G. Palermo

Dynamic Systems Inc.

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THERMOCOUPLE MEASUREMENT SYSTEM

DEGREES C X DEGREES F

REFERENCE TEMPERATURE INPUT		TC- 3 TYPE K			TC- 4 TYPE R		
DEG. C	DEG. F	AS FOUND	OOT	AFTER CALIBRATION	AS FOUND	OOT	AFTER CALIBRATION
200	400	200		200	***		***
300	600	300		300	***		***
400	800	400		400	***		***
500	1000	500		500	500		500
600	1200	600		600	600		600
700	1400	700		700	700		700
800	1600	800		800	800		800
900	1800	9000		9000	900		900
1000	2000	1000		1000	1000		1000
1100	2200	1100		1100	1100		1100
1200	2400	1200		1200	1200		1200
1300	2600	1300		1300	1300		1300
1400	2800	****		****	1400		1400
1500	3000	****		****	1500		1500
1600	3200	****		****	1600		1600
1700	3400	****		****	1700		1700

Note: Platinum Thermocouples - start @ 500°C (932°F)
Chromel-Alumel Thermocouples - start @ 200°C (392°F)

Date of Calibration: 2/28/2022

Calibrated by: G. Palermo

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STROKE CALIBRATION

REFERENCE INSTRUMENT	MODEL NUMBER	SERIAL NUMBER	CALIBRATION DUE DATE
Mitutoyo	0-6”	1300046	7/28/2022

DEVICE TYPE BEING CALIBRATED		SERIAL NO. 130911B24558				
TYPE		RANGE			MOUNTING	
X	Stroke	X	+/- 60.00	mm	X	Installed
	Diametrial Extensometer			in.		Free Standing
	Lengthwise Extensometer			kN		In Calibration Fixture
	Load Cell			N		
				kg		
				lbs.		

CALIBRATION IN INCHES ____ OR MILLIMETERS X

REFERENCE LENGTH	AS FOUND	OOT	AFTER CALIBRATION
-60.00	-60.09		-60.09
-50.00	-50.03		-50.03
-40.00	-40.09		-40.09
-30.00	-30.12		-30.12
-20.00	-20.06		-20.06
-10.00	-10.07		-10.07
0.00	0.00		0.00
10.00	10.05		10.05
20.00	20.10		20.10
30.00	30.04		30.04
40.00	40.05		40.05
50.00	49.97		49.97
60.00	60.00		60.00

Date of Calibration: 2/28/2022

Calibrated by: G. Palermo

Dynamic Systems Inc.

CERTIFICATE OF CALIBRATION

Customer: University of Witwatersrand Serial No. 504

STANDARD JAW FORCE SYSTEM CALIBRATION

REFERENCE INSTRUMENT	SSI SERIAL #	LOAD CELL SERIAL #	CALIBRATION DUE DATE
Trandsucer Techniques	232342	120626	6/17/2022

DEVICE TYPE BEING CALIBRATED SERIAL NO.						
TYPE		RANGE			MOUNTING	
	Stroke			mm	X	Installed
	Diametrial Extensometer			in.		Free Standing
	Lengthwise Extensometer	X	-/+111kN	kN		In Calibration Fixture
X	Load Cell			N		
				Kg		
				lbs.		

CALIBRATION IN KILONEWTONS X NEWTONS ____ KILOGRAMS ____ POUNDS ____

	TENSION			COMPRESSION		
REFERENCE LOAD	AS FOUND	OOT	AFTER CALIBRATION	AS FOUND	OOT	AFTER CALIBRATION
0.00	0.00		0.00	0.00		0.00
10.00	9.96		9.96	-10.26		-10.26
20.00	19.93		19.93	-20.36		-20.36
30.00	29.93		29.93	-30.44		-30.44
40.00	39.81		39.81	-40.55		-40.55
50.00	49.68		49.68	-50.65		-50.65
60.00	59.55		59.55	-60.74		-60.74
70.00	69.41		69.41	-70.84		-70.84
80.00	79.29		79.29	-80.92		-80.92

Date of Calibration: 2/28/2022

Calibrated by: G. Palermo

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C-STRAIN EXTENSOMETER CALIBRATION

REFERENCE INSTRUMENT	MODEL NUMBER	SERIAL NUMBER	CALIBRATION DUE DATE
Mitutoyo	ID Micrometer	71259725	9/13/2022

DEVICE TYPE BEING CALIBRATED		SERIAL NO. J51797				
TYPE		RANGE			MOUNTING	
	Stroke	X	+/-6.35	mm		Installed
X	Diametrial Extensometer			in.		Free Standing
	Lengthwise Extensometer			kN	X	In Calibration Fixture
	Load Cell			N		
				kg		
				lbs.		

CALIBRATION IN INCHES 0.1 ___ OR MILLIMETERS X ___

REFERENCE DIAMETER	AS FOUND	OOT	AFTER CALIBRATION
-6.35	-6.35		-6.35
-5.00	-5.01		-5.01
-4.00	-4.02		-4.02
-3.00	-3.01		-3.01
-2.00	-2.01		-2.01
-1.00	-1.00		-1.00
0.00	0.00		0.00
1.00	0.98		0.98
2.00	1.99		1.99
3.00	2.98		2.98
4.00	4.00		4.00
5.00	4.99		4.99
6.35	6.34		6.34

Date of Calibration: 2/28/2022

Calibrated by: G. Palermo

Dynamic Systems Inc.

CERTIFICATE OF CALIBRATION

Customer: University of Witwatersrand Serial No. 504

CCT DILATOMETER CALIBRATION - 39018

REFERENCE INSTRUMENT	MODEL NUMBER	SERIAL NUMBER	CALIBRATION DUE DATE
Mitutoyo	ID Micrometer	71259725	9/13/2022

DEVICE TYPE BEING CALIBRATED		SERIAL NO. J57032				
TYPE		RANGE			MOUNTING	
	Stroke	X	+/-2.50	mm		Installed
X	Diametrial Extensometer			in.		Free Standing
	Lengthwise Extensometer			kN	X	In Calibration Fixture
	Load Cell			N		
				kg		
				lbs.		

CALIBRATION IN 0.1 INCHES ____ OR 0.1 MILLIMETERS X ____

REFERENCE DIAMETER	AS FOUND	OOT	AFTER CALIBRATION
-2.50	-2.49		-2.49
-2.00	-2.00		-2.00
-1.50	-2.50		-2.50
-1.00	-0.99		-0.99
-0.50	-0.50		-0.50
0.00	0.00		0.00
0.50	0.50		0.50
1.00	1.01		1.01
1.50	1.51		1.51
2.00	2.01		2.01
2.50	2.50		2.50

Date of Calibration: 2/28/2022

Calibrated by: G. Palermo

Dynamic Systems Inc.

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HOT ZONE L-STRAIN - HZT071

REFERENCE INSTRUMENT	MODEL NUMBER	SERIAL NUMBER	CALIBRATION DUE DATE
Mitutoyo	ID Micrometer	71259725	9/13/2022

DEVICE TYPE BEING CALIBRATED SERIAL NO. E91773						
TYPE		RANGE			MOUNTING	
	Stroke	X	+5/-2	mm		Installed
	Diametrial Extensometer			in.		Free Standing
X	Lengthwise Extensometer			kN	X	In Calibration Fixture
	Load Cell			N		
				kg		
				lbs.		

CALIBRATION IN 0.1 INCHES ____ OR MILLIMETERS X ____

MEASURED DISTANCE	AS FOUND	OOT	AFTER CALIBRATION
-2.00	-2.00		-2.00
-1.50	-1.51		-1.51
-1.00	-1.01		-1.01
-0.50	-0.51		-0.51
0.00	0.00		0.00
1.00	0.96		0.96
2.00	1.97		1.97
3.00	2.99		2.99
4.00	4.00		4.00
5.00	5.01		5.01

Date of Calibration: 2/28/2022

Calibrated by: G. Palermo